



IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

Institut Universitaire des sciences

- Faculté des sciences et de technologie

✓ TD N°5 Modélisation

Nom & Prénom : COFFY Cliford

Niveau : L3

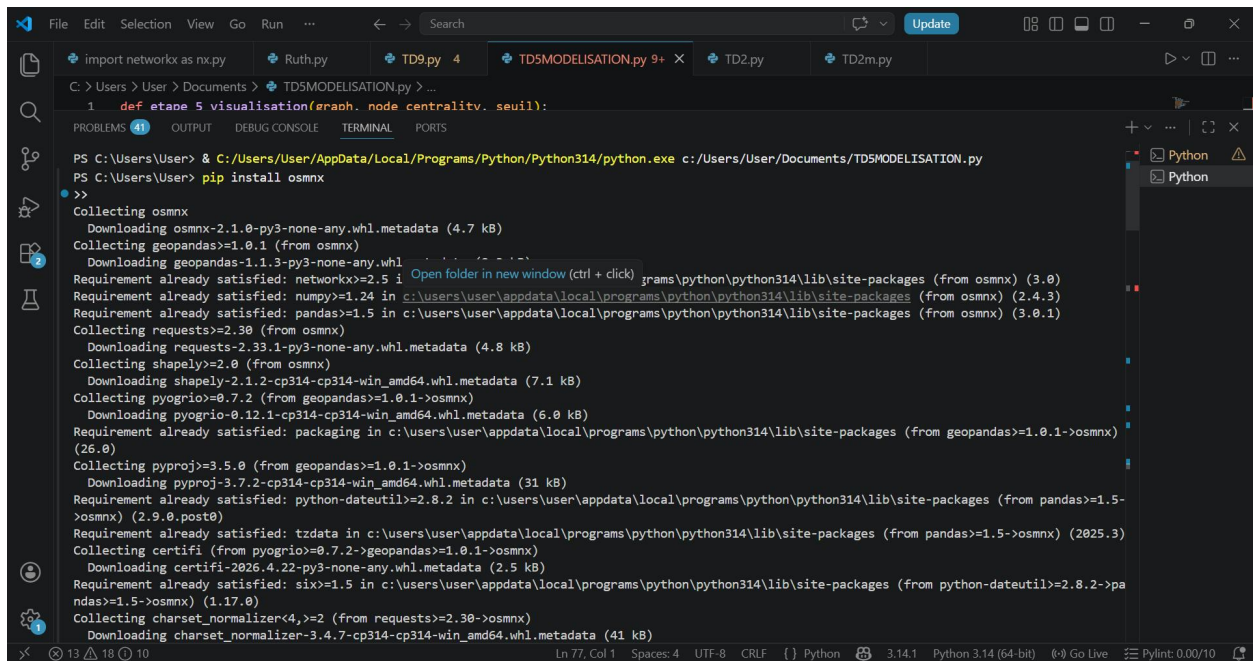
Prof. : Ismaël SAINT AMOUR

Date: 08/05/2026

TD1 - Analyse du réseau routier de Cap-Haïtien avec OpenStreetMap

Travail demandé

1. Charger la carte
2. Nettoyer
3. Calculer métriques
4. Identifier points critiques
5. Visualisation avancée



```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py 9+ X TD2.py TD2m.py
C:\Users\User> Documents > TD5MODELISATION.py > ...
1 def etape 5 visualisation(eraoh, node centralitv, seuil):
PROBLEMS 41 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User> & C:/Users/User/AppData/Local/Programs/Python/Python314/python.exe c:/Users/User/Documents/TD5MODELISATION.py
PS C:\Users\User> pip install osmnx
>>>
Collecting osmnx
  Downloading osmnx-2.1.0-py3-none-any.whl.metadata (4.7 kB)
Collecting geopandas>=1.0.1 (from osmnx)
  Downloading geopandas-1.1.3-py3-none-any.whl.metadata (1.9 kB)
Requirement already satisfied: networkx>=2.5 in c:\users\user\appdata\local\programs\python\python314\lib\site-packages (from osmnx) (3.0)
Requirement already satisfied: numpy>=1.24 in c:\users\user\appdata\local\programs\python\python314\lib\site-packages (from osmnx) (2.4.3)
Requirement already satisfied: pandas>=1.5 in c:\users\user\appdata\local\programs\python\python314\lib\site-packages (from osmnx) (3.0.1)
Collecting requests>=2.30 (from osmnx)
  Downloading requests-2.33.1-py3-none-any.whl.metadata (4.8 kB)
Collecting shapely>=2.0 (from osmnx)
  Downloading shapely-2.1.2-cp314-cp314-win_amd64.whl.metadata (7.1 kB)
Collecting pyogrio>=0.7.2 (from geopandas>=1.0.1->osmnx)
  Downloading pyogrio-0.12.1-cp314-cp314-win_amd64.whl.metadata (6.0 kB)
Requirement already satisfied: packaging in c:\users\user\appdata\local\programs\python\python314\lib\site-packages (from geopandas>=1.0.1->osmnx) (26.0)
Collecting pyproj>=3.5.0 (from geopandas>=1.0.1->osmnx)
  Downloading pyproj-3.7.2-cp314-cp314-win_amd64.whl.metadata (31 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\user\appdata\local\programs\python\python314\lib\site-packages (from pandas>=1.5->osmnx) (2.9.0.post0)
Requirement already satisfied: tzdata in c:\users\user\appdata\local\programs\python\python314\lib\site-packages (from pandas>=1.5->osmnx) (2025.3)
Collecting certifi (from pyogrio>=0.7.2->geopandas>=1.0.1->osmnx)
  Downloading certifi-2026.4.22-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: six>=1.5 in c:\users\user\appdata\local\programs\python\python314\lib\site-packages (from python-dateutil>=2.8.2->pandas>=1.5->osmnx) (1.17.0)
Collecting charset-normalizer<4,>=2 (from requests>=2.30->osmnx)
  Downloading charset_normalizer-3.4.7-cp314-cp314-win_amd64.whl.metadata (41 kB)
Ln 77, Col 1 Spaces: 4 UTF-8 CRLF Python 3.14.1 Python 3.14 (64-bit) Go Live Pylint: 0.00/10
```

```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py 9+ X TD2.py TD2m.py
C:\Users\User> Documents > TD5MODELISATION.py > ...
1 def etape 5 visualisation(graph, node centrality, seuil):
PROBLEMS 41 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User> pip install osmnx
>>
Downloading certifi-2026.4.22-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: six>=1.5 in c:\users\user\appdata\local\programs\python\python314\lib\site-packages (from python-dateutil>=2.8.2->pa
ndas>=1.5->osmnx) (1.17.0)
Collecting charset_normalizer<4,>=2 (from requests>=2.30->osmnx)
Downloading charset_normalizer-3.4.7-cp314-cp314-win_amd64.whl.metadata (41 kB)
Collecting idna<4,>=2.5 (from requests>=2.30->osmnx)
Downloading idna-3.13-py3-none-any.whl.metadata (8.0 kB)
Collecting urllib3<3,>=1.26 (from requests>=2.30->osmnx)
Downloading urllib3-2.7.0-py3-none-any.whl.metadata (6.9 kB)
Downloading osmnx-2.1.0-py3-none-any.whl (104 kB)
Downloading geopandas-1.1.3-py3-none-any.whl (342 kB)
Downloading pyogrio-0.12.1-cp314-cp314-win_amd64.whl (23.6 MB)
23.6/23.6 MB 443.8 kB/s 0:01:05
Downloading pyproj-3.7.2-cp314-cp314-win_amd64.whl (6.4 MB)
6.4/6.4 MB 864.0 kB/s 0:00:07
Downloading requests-2.33.1-py3-none-any.whl (64 kB)
Downloading charset_normalizer-3.4.7-cp314-cp314-win_amd64.whl (159 kB)
Downloading idna-3.13-py3-none-any.whl (68 kB)
Downloading urllib3-2.7.0-py3-none-any.whl (131 kB)
Downloading certifi-2026.4.22-py3-none-any.whl (135 kB)
Downloading shapely-2.1.2-cp314-cp314-win_amd64.whl (1.8 MB)
1.8/1.8 MB 152.8 kB/s 0:00:11
Installing collected packages: urllib3, shapely, idna, charset_normalizer, certifi, requests, pyproj, pyogrio, geopandas, osmnx
Successfully installed certifi-2026.4.22 charset_normalizer-3.4.7 geopandas-1.1.3 idna-3.13 osmnx-2.1.0 pyogrio-0.12.1 pyproj-3.7.2 requests-2.33.1
shapely-2.1.2 urllib3-2.7.0
[notice] A new release of pip is available: 25.3 -> 26.1
Ln 77, Col 1 Spaces: 4 UTF-8 CRLF {} Python 3.14.1 Python 3.14 (64-bit) Go Live Pylint: 0.00/10
```

```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py 9+ X TD2.py TD2m.py
C:\Users\User> Documents > TD5MODELISATION.py > ...
1 def etape 5 visualisation(graph, node centrality, seuil):
PROBLEMS 41 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User> & C:/Users/User/AppData/Local/Programs/Python/Python314/python.exe c:/Users/User/Documents/TD5MODELISATION.py
KeyboardInterrupt
PS C:\Users\User> python -m pip install --upgrade pip setuptools wheel
>>
Requirement already satisfied: pip in c:\users\user\appdata\local\programs\python\python314\lib\site-packages (25.3)
Collecting pip
Downloading pip-26.1.1-py3-none-any.whl.metadata (4.6 kB)
Collecting setuptools
Downloading setuptools-82.0.1-py3-none-any.whl.metadata (6.5 kB)
Collecting wheel
Downloading wheel-0.47.0-py3-none-any.whl.metadata (2.3 kB)
Requirement already satisfied: packaging>=24.0 in c:\users\user\appdata\local\programs\python\python314\lib\site-packages (from wheel) (26.0)
Downloading pip-26.1.1-py3-none-any.whl (1.8 MB)
1.8/1.8 MB 2.4 MB/s 0:00:00
Downloading setuptools-82.0.1-py3-none-any.whl (1.0 MB)
1.0/1.0 MB 2.5 MB/s 0:00:00
Downloading wheel-0.47.0-py3-none-any.whl (32 kB)
Installing collected packages: wheel, setuptools, pip
Attempting uninstall: pip
Found existing installation: pip 25.3
Uninstalling pip-25.3:
Successfully uninstalled pip-25.3
Successfully installed pip-26.1.1 setuptools-82.0.1 wheel-0.47.0
PS C:\Users\User> pip uninstall scipy scikit-learn -y
Found existing installation: scipy 1.17.1
Uninstalling scipy-1.17.1:
Successfully uninstalled scipy-1.17.1
Found existing installation: scikit-learn 1.8.0
Uninstalling scikit-learn-1.8.0:
Ln 77, Col 1 Spaces: 4 UTF-8 CRLF {} Python 3.14.1 Python 3.14 (64-bit) Go Live Pylint: 0.00/10
```

```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py 3 X TD2.py TD2m.py
C: > Users > User > Documents > TD5MODELISATION.py > ...
1 import osmnx as ox
2 import networkx as nx
3 import matplotlib.pyplot as plt
4 import matplotlib.patches as mpatches
5 import numpy as np
6 import sys
7 import time
8
9 ox.settings.log_console = False
10 ox.settings.use_cache = True
11
12 def separateur():
13     print("=" * 55)
14
15 def pause():
16     time.sleep(0.6)
17
18 def afficher_menu():
19     separateur()
20     print("  ANALYSE DU RESEAU ROUTIER --- CAP-HAITIEN")
21     separateur()
22     print("")
23     Veuillez choisir les etapes a executer.
24     Vous pouvez entrer les numeros un par un.
```

```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py 3 X TD2.py TD2m.py
C: > Users > User > Documents > TD5MODELISATION.py > ...
94 def etape_3_metriques(graph):
95     print("  calcul de la centralite en cours...")
96     graphe_non_oriente = graph.to_undirected()
97     node_centrality = nx.betweenness centrality(graphe_non_oriente)
98     nx.set_node_attributes(graph, node_centrality, 'betweenness')
99     print("  Centralite calculee.")
100     return graph, node_centrality
101
102 def etape_4_points_critiques(graph, node_centrality):
103     separateur()
104     print("  ETAPE 4 : Identification des points critiques")
105     separateur()
106     pause()
107     seuil = np.percentile(list(node_centrality.values()), 90)
108     points_critiques = {n: v for n, v in node_centrality.items() if v >= seuil}
109
110     stats = ox.basic_stats(graph)
111     print(f"  {len(points_critiques)} points critiques identifiés (top 10% centralite).")
112     print(f"  Centralite maximale : {max(node_centrality.values()):.6f}")
113     print(f"  Centralite moyenne : {np.mean(list(node_centrality.values())):.6f}")
114     print(f"  Noeuds : {stats['n']}")
115     print(f"  Aretes : {stats['m']}")
116     print(f"  Longueur totale : {stats['edge_length_total']/1000:.2f} km")
```

```
import networkx as nx.py
Ruth.py
TD9.py 4
TD5MODELISATION.py 3 X
TD2.py
TD2m.py

C: > Users > User > Documents > TD5MODELISATION.py > ...

234
240 def main():
241     if graph is None:
242         graph = etape_1_charger()
243         graph = etape_2_nettoyer(graph)
244         graph, node centrality = etape_3_metriques(graph)
245         seuil = etape_4_points_critiques(graph, node centrality)
246
247     if 5 in etapes:
248         if not node centrality or seuil == 0:
249             print(" Les etapes 3 et 4 sont requises. Calcul automatique...")
250             if graph is None:
251                 graph = etape_1_charger()
252                 graph = etape_2_nettoyer(graph)
253                 graph, node centrality = etape_3_metriques(graph)
254                 seuil = etape_4_points_critiques(graph, node centrality)
255             etape_5_visualisation(graph, node centrality, seuil)
256
257     message_au_revoir(etapes)
258
259 main()

PROBLEMS 29 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User> & C:\Users\User\AppData\Local\Programs\Python\Python314\python.exe c:\Users\User\Documents\TD5MODELISATION.py
=====
Affichage de la carte en cours...
```

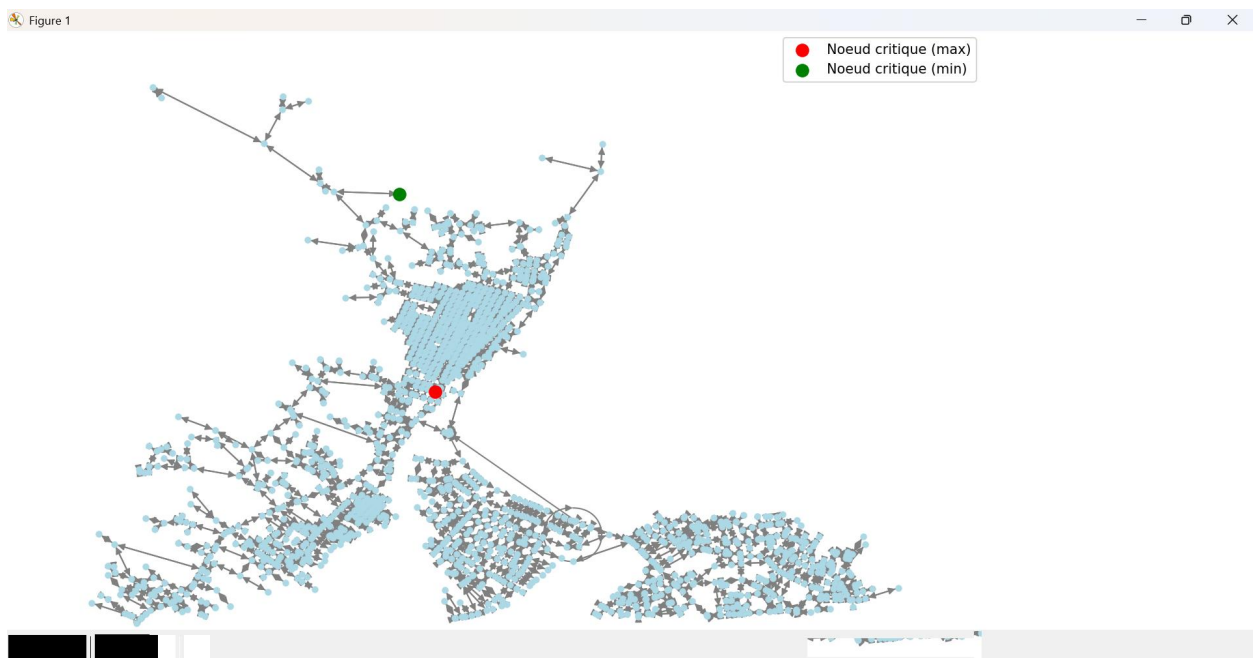
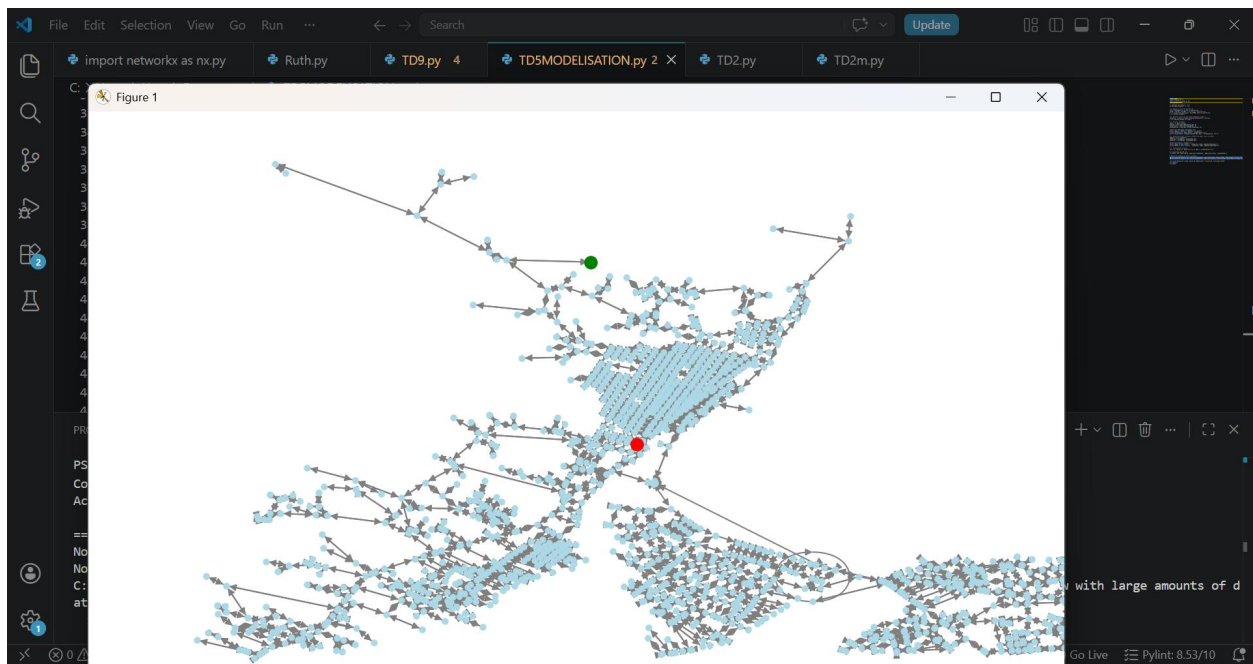
```
import networkx as nx.py
Ruth.py
TD9.py 4
TD5MODELISATION.py 9+ X
TD2.py
TD2m.py

C: > Users > User > Documents > TD5MODELISATION.py > ...

1 def etape_5_visualisation(graph, node centrality, seuil):
2     print("ANALYSE DU RESEAU ROUTIER --- CAP-HAITIEN")
3     print("=====")
4     print("Veuillez choisir les etapes a executer.")
5     print("Vous pouvez entrer les numeros un par un.")
6     print("Tapez 0 pour lancer l'analyse avec vos choix.")
7     print("Tapez 9 pour tout selectionner.")
8
9     [1] Charger la carte OpenStreetMap
10    [2] Nettoyer le graphe
11    [3] Calculer les metriques (vitesse + temps)
12    [4] Identifier les points critiques
13    [5] Visualisation avancee
14    [0] Lancer l'analyse
15    [9] Tout selectionner
16
17    =====
18    Entrez un numero : 1
19    Etape 1 ajoutee. Etapes selectionnees : [1]
20    Entrez un numero : 2
21    Etape 2 ajoutee. Etapes selectionnees : [1, 2]
22    Entrez un numero : 3
23    Etape 3 ajoutee. Etapes selectionnees : [1, 2, 3]
24    Entrez un numero : 4
25    Etape 4 ajoutee. Etapes selectionnees : [1, 2, 3, 4]
26    Entrez un numero : 5
27    Etape 5 ajoutee. Etapes selectionnees : [1, 2, 3, 4, 5]
```

```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py 9+ X TD2.py TD2m.py
C:\Users\User> Documents > TD5MODELISATION.py > ...
1 def etape 5 visualisation(eraah. node centralitv. seuil):
PROBLEMS 41 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User> & C:/Users/User/AppData/Local/Programs/Python/Python314/python.exe c:/Users/User/Documents/TD5MODELISATION.py
Entrez un numero : 1
Etape 1 ajoutée. Etapes selectionnees : [1]
Entrez un numero : 2
Etape 2 ajoutée. Etapes selectionnees : [1, 2]
Entrez un numero : 3
Etape 3 ajoutée. Etapes selectionnees : [1, 2, 3]
Entrez un numero : 4
Etape 4 ajoutée. Etapes selectionnees : [1, 2, 3, 4]
Entrez un numero : 5
Etape 5 ajoutée. Etapes selectionnees : [1, 2, 3, 4, 5]
Entrez un numero : 0
=====
ETAPE 1 : Chargement de la carte OpenStreetMap
=====
Carte chargée avec succes.
=====
ETAPE 2 : Nettoyage du graphe
=====
Graphe nettoye --- 1409 noeuds | 3970 aretes conserves.
=====
ETAPE 3 : Calcul des metriques de vitesse et de temps
=====
Vitesses et temps de trajet calcules.
Calcul de la centralite en cours...
Centralite calculee.
=====
ETAPE 4 : Identification des points critiques
=====
```

```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py 9+ X TD2.py TD2m.py
C:\Users\User> Documents > TD5MODELISATION.py > ...
1 def etape 5 visualisation(eraah. node centralitv. seuil):
PROBLEMS 41 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User> & C:/Users/User/AppData/Local/Programs/Python/Python314/python.exe c:/Users/User/Documents/TD5MODELISATION.py
Entrez un numero : 0
=====
ETAPE 1 : Chargement de la carte OpenStreetMap
=====
Carte chargée avec succes.
=====
ETAPE 2 : Nettoyage du graphe
=====
Graphe nettoye --- 1409 noeuds | 3970 aretes conserves.
=====
ETAPE 3 : Calcul des metriques de vitesse et de temps
=====
Vitesses et temps de trajet calcules.
Calcul de la centralite en cours...
Centralite calculee.
=====
ETAPE 4 : Identification des points critiques
=====
141 points critiques identifiées (top 10% centralite).
Centralite maximale : 0.434913
Centralite moyenne : 0.020907
Noeuds : 1409
Aretes : 3970
Longueur totale : 328.58 km
=====
ETAPE 5 : Visualisation avancee
=====
Affichage de la carte en cours...
```

TD2 - Optimisation du transport urbain à Port-au-Prince avec OSMnx

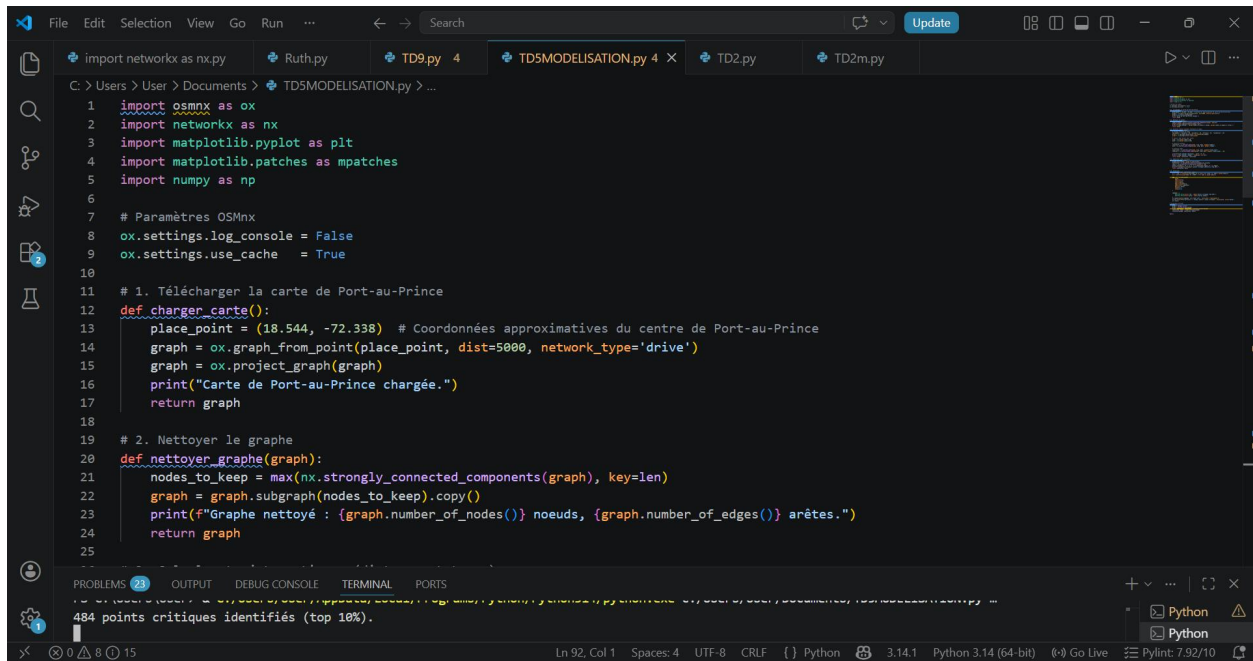
Travail demandé

1. Télécharger la carte
2. Nettoyer le graphe
3. Calculer trajets optimaux

Dijkstra (distance)

temps estimé

4. Identifier zones critiques
5. Visualiser



```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py x TD2.py TD2m.py
C:\Users\User\Documents> TD5MODELISATION.py > ...
1 import osmnx as ox
2 import networkx as nx
3 import matplotlib.pyplot as plt
4 import matplotlib.patches as mpatches
5 import numpy as np
6
7 # Paramètres OSMnx
8 ox.settings.log_console = False
9 ox.settings.use_cache = True
10
11 # 1. Télécharger la carte de Port-au-Prince
12 def charger_carte():
13     place_point = (18.544, -72.338) # Coordonnées approximatives du centre de Port-au-Prince
14     graph = ox.graph_from_point(place_point, dist=5000, network_type='drive')
15     graph = ox.project_graph(graph)
16     print("Carte de Port-au-Prince chargée.")
17     return graph
18
19 # 2. Nettoyer le graphe
20 def nettoyer_graphe(graph):
21     nodes_to_keep = max(nx.strongly_connected_components(graph), key=len)
22     graph = graph.subgraph(nodes_to_keep).copy()
23     print(f"Graphe nettoyé : {graph.number_of_nodes()} noeuds, {graph.number_of_edges()} arêtes.")
24     return graph
25
PROBLEMS 25 OUTPUT DEBUG CONSOLE TERMINAL PORTS
484 points critiques identifiés (top 10%).
Python Python
Ln 92, Col 1 Spaces: 4 UTF-8 CRLF Python 3.14.1 Python 3.14 (64-bit) Go Live Pylint: 7.92/10
```

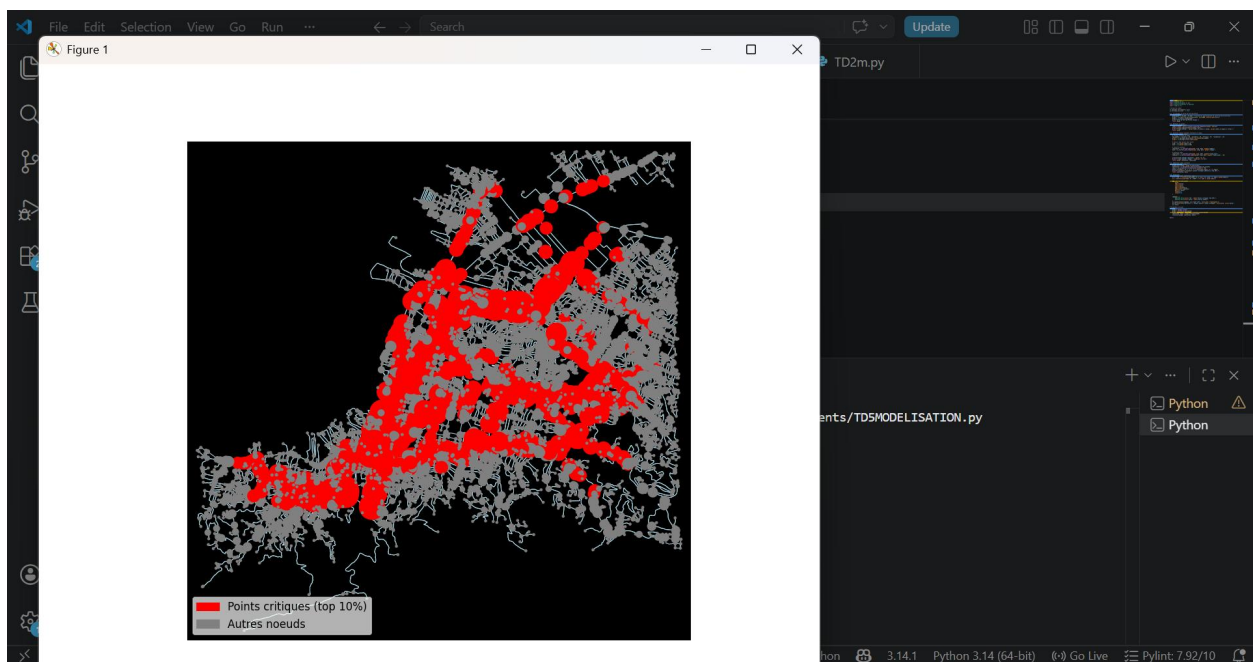


```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py X TD2.py TD2m.py
C:\Users\User> Documents > TD5MODELISATION.py > ...
1 import osmnx as ox
2 import networkx as nx
3 import matplotlib.pyplot as plt
4 import matplotlib.patches as mpatches
5 import numpy as np
6
7 # Paramètres OSMnx
8 ox.settings.log_console = False
9 ox.settings.use_cache = True
10
11 # 1. Télécharger la carte de Port-au-Prince
12 def charger_carte():
13     place_point = (18.544, -72.338) # Coordonnées approximatives du centre de Port-au-Prince
14     graph = ox.graph_from_point(place_point, dist=5000, network_type='drive')
15     graph = ox.project_graph(graph)
16     print("Carte de Port-au-Prince chargée.")
17     return graph
18
19 # 2. Nettoyer le graphe
20 def nettoyer_graphe(graph):
21     nodes_to_keep = max(nx.strongly_connected_components(graph), key=len)
22     graph = graph.subgraph(nodes_to_keep).copy()
23     print(f"Graphe nettoyé : {graph.number_of_nodes()} noeuds, {graph.number_of_edges()} arêtes.")
24     return graph
25
PROBLEMS 23 OUTPUT DEBUG CONSOLE TERMINAL PORTS
484 points critiques identifiés (top 10%).
Python Python
Ln 92, Col 1 Spaces: 4 UTF-8 CRLF {} Python 3.14.1 Python 3.14 (64-bit) Go Live Pylint: 7.92/10
```

```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py X TD2.py TD2m.py
C:\Users\User> Documents > TD5MODELISATION.py > ...
59 def visualiser(graph, centrality, seuil):
60     close=False
61
62     legende = [
63         mpatches.Patch(color='red', label='Points critiques (top 10%)'),
64         mpatches.Patch(color='gray', label='Autres noeuds')
65     ]
66     ax.legend(handles=legende, loc='lower left', fontsize=8, framealpha=0.7)
67     ax.set_title("Port-au-Prince --- Réseau routier & zones critiques", fontsize=14, color='white')
68     plt.show()
69
70 # Programme principal
71 def main():
72     graph = charger_carte()
73     graph = nettoyer_graphe(graph)
74     graph, route_dist, route_time = calculer_trajets(graph)
75     centrality, seuil = zones_critiques(graph)
76     visualiser(graph, centrality, seuil)
77
78 main()
79
PROBLEMS 23 OUTPUT DEBUG CONSOLE TERMINAL PORTS
484 points critiques identifiés (top 10%).
Python Python
Ln 92, Col 1 Spaces: 4 UTF-8 CRLF {} Python 3.14.1 Python 3.14 (64-bit) Go Live Pylint: 7.92/10
```

```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py x TD2.py TD2m.py
C: > Users > User > Documents > TD5MODELISATION.py > ...
84 def main():
88     centrality, seuil = zones_critiques(graph)
89     visualiser(graph, centrality, seuil)
90
91 main()
92

PROBLEMS 23 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User> & C:/Users/User/AppData/Local/Programs/Python/Python314/python.exe c:/Users/User/Documents/TD5MODELISATION.py
Carte de Port-au-Prince chargée.
Graphe nettoyé : 4832 noeuds, 12365 arêtes.
Trajet optimal (distance) : 4572.79 m
Trajet optimal (temps) : 8.11 min
484 points critiques identifiés (top 10%).
```





TD3 - Optimisation des transports (tap-tap)

Objectif

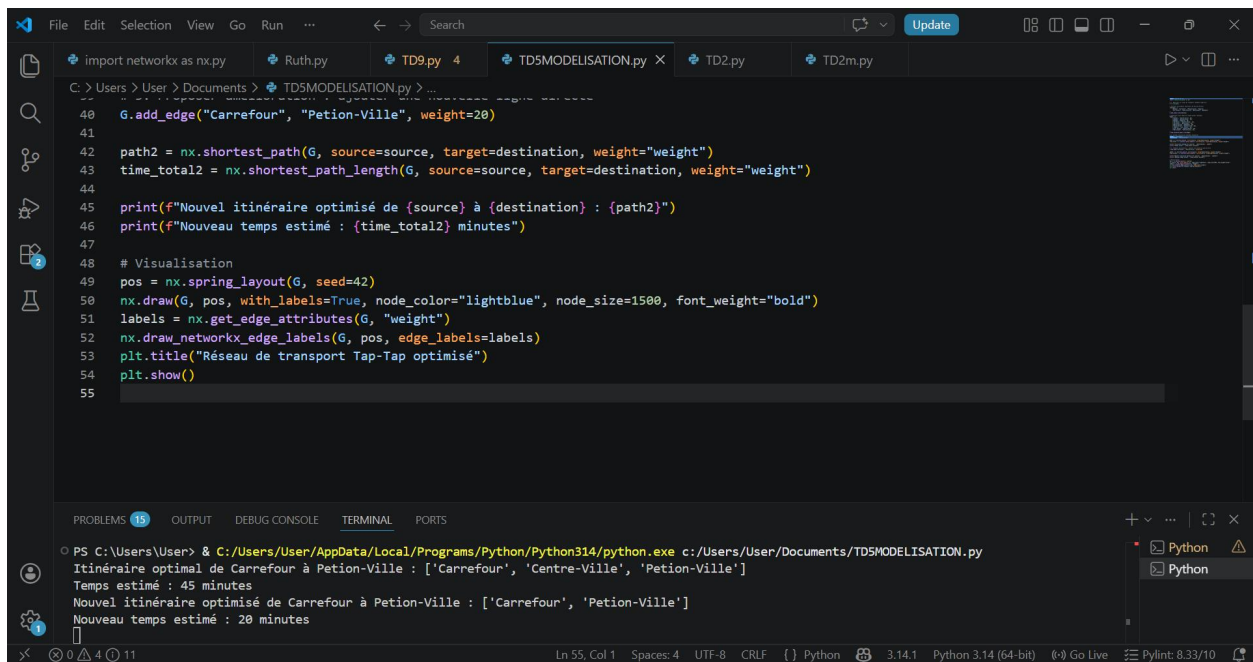
Améliorer les trajets urbains.

Travail demandé

1. Modéliser un réseau de transport
2. Trouver itinéraires optimaux
3. Proposer amélioration

Résultat attendu

réseau optimisé



```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py X TD2.py TD2m.py
C:\Users\User> Documents > TD5MODELISATION.py > ...
40 G.add_edge("Carrefour", "Petion-Ville", weight=20)
41
42 path2 = nx.shortest_path(G, source=source, target=destination, weight="weight")
43 time_total2 = nx.shortest_path_length(G, source=source, target=destination, weight="weight")
44
45 print(f"Nouvel itinéraire optimisé de {source} à {destination} : {path2}")
46 print(f"Nouveau temps estimé : {time_total2} minutes")
47
48 # Visualisation
49 pos = nx.spring_layout(G, seed=42)
50 nx.draw(G, pos, with_labels=True, node_color="lightblue", node_size=1500, font_weight="bold")
51 labels = nx.get_edge_attributes(G, "weight")
52 nx.draw_networkx_edge_labels(G, pos, edge_labels=labels)
53 plt.title("Réseau de transport Tap-Tap optimisé")
54 plt.show()
55
```

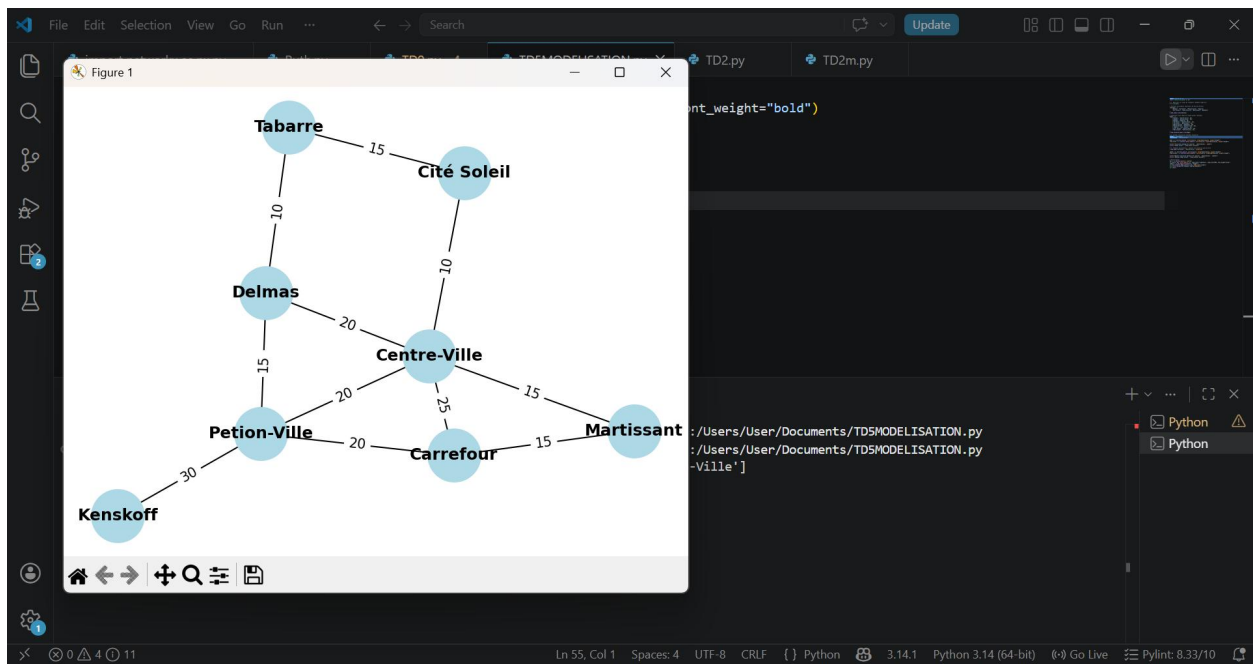
PROBLEMS 15 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\User> & C:/Users/User/AppData/Local/Programs/Python/Python314/python.exe c:/Users/User/Documents/TD5MODELISATION.py
Itinéraire optimal de Carrefour à Petion-Ville : ['Carrefour', 'Centre-Ville', 'Petion-Ville']
Temps estimé : 45 minutes
Nouvel itinéraire optimisé de Carrefour à Petion-Ville : ['Carrefour', 'Petion-Ville']
Nouveau temps estimé : 20 minutes
```

Ln 55, Col 1 Spaces: 4 UTF-8 CRLF Python 3.14.1 Python 3.14 (64-bit) Go Live Pylint: 8.33/10

```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py X TD2.py TD2m.py
C: > Users > User > Documents > TD5MODELISATION.py > ...
1 import networkx as nx
2 import matplotlib.pyplot as plt
3
4 # 1. Modéliser un réseau de transport (exemple simplifié)
5 G = nx.Graph()
6
7 # Stations principales (quartiers de Port-au-Prince)
8 stations = [
9     "Delmas", "Carrefour", "Petion-Ville", "Tabarre",
10    "Cité Soleil", "Centre-Ville", "Martissant", "Kenscoff"
11 ]
12 G.add_nodes_from(stations)
13
14 # Connexions avec temps de trajet estimé (minutes)
15 edges = [
16     ("Delmas", "Centre-Ville", 20),
17     ("Delmas", "Petion-Ville", 15),
18     ("Delmas", "Tabarre", 10),
19     ("Carrefour", "Centre-Ville", 25),
20     ("Carrefour", "Martissant", 15),
21     ("Petion-Ville", "Kenscoff", 30),
22     ("Petion-Ville", "Centre-Ville", 20).
23 ]
24
25 PROBLEMS 15 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User> & C:/Users/User/AppData/Local/Programs/Python/Python314/python.exe c:/Users/User/Documents/TD5MODELISATION.py
Itinéraire optimal de Carrefour à Petion-Ville : ['Carrefour', 'Centre-Ville', 'Petion-Ville']
Temps estimé : 45 minutes
Nouvel itinéraire optimisé de Carrefour à Petion-Ville : ['Carrefour', 'Petion-Ville']
Nouveau temps estimé : 20 minutes
Ln 55, Col 1 Spaces: 4 UTF-8 CRLF {} Python 3.14.1 Python 3.14 (64-bit) Go Live Pylint: 8.33/10
```

```
File Edit Selection View Go Run ... Search Update
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py X TD2.py TD2m.py
C: > Users > User > Documents > TD5MODELISATION.py > ...
50 nx.draw(G, pos, with_labels=True, node_color="lightblue", node_size=1500, font_weight="bold")
51 labels = nx.get_edge_attributes(G, "weight")
52 nx.draw_networkx_edge_labels(G, pos, edge_labels=labels)
53 plt.title("Réseau de transport Tap-Tap optimisé")
54 plt.show()
55
PROBLEMS 15 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User> & C:/Users/User/AppData/Local/Programs/Python/Python314/python.exe c:/Users/User/Documents/TD5MODELISATION.py
PS C:\Users\User> & C:/Users/User/AppData/Local/Programs/Python/Python314/python.exe c:/Users/User/Documents/TD5MODELISATION.py
Itinéraire optimal de Carrefour à Petion-Ville : ['Carrefour', 'Centre-Ville', 'Petion-Ville']
Temps estimé : 45 minutes
Nouvel itinéraire optimisé de Carrefour à Petion-Ville : ['Carrefour', 'Petion-Ville']
Nouveau temps estimé : 20 minutes
Ln 55, Col 1 Spaces: 4 UTF-8 CRLF {} Python 3.14.1 Python 3.14 (64-bit) Go Live Pylint: 8.33/10
```



TD 4 - Comparaison entre villes

Objectif

Comparer plusieurs villes d'Haïti.

Travail demandé

Comparer :

Port-au-Prince

Cap-Haïtien

Les Cayes

Jacmel

Port-de-paix

Critères :

densité

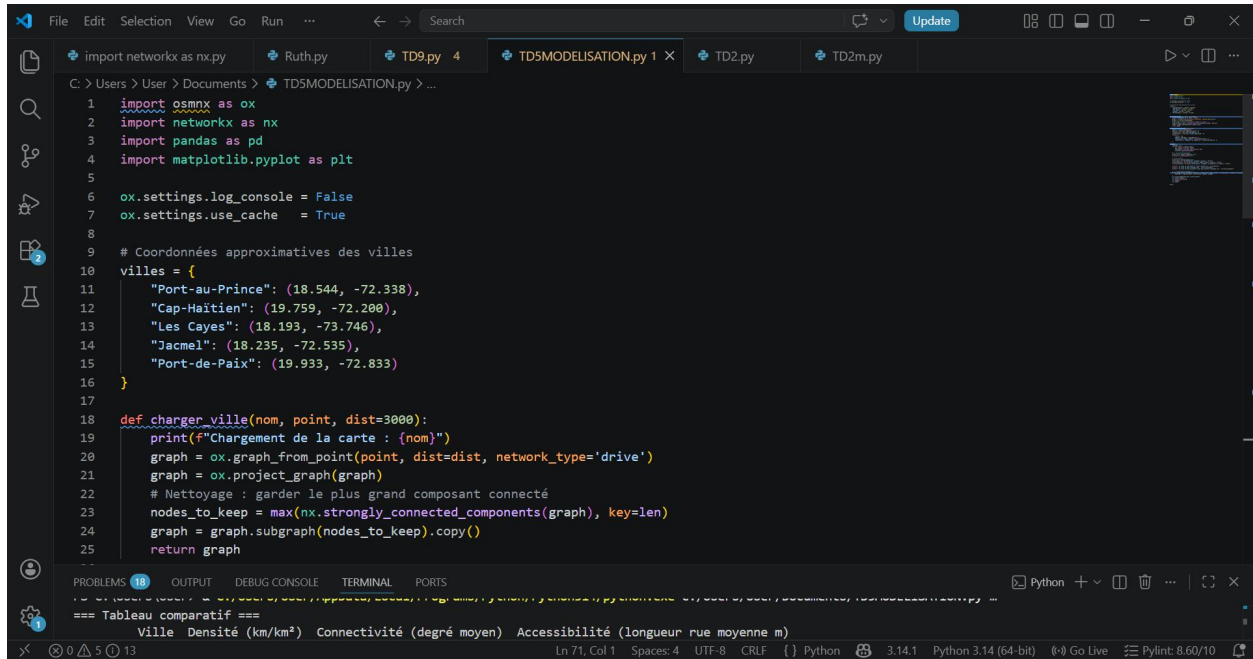
connectivité

accessibilité

Résultat attendu

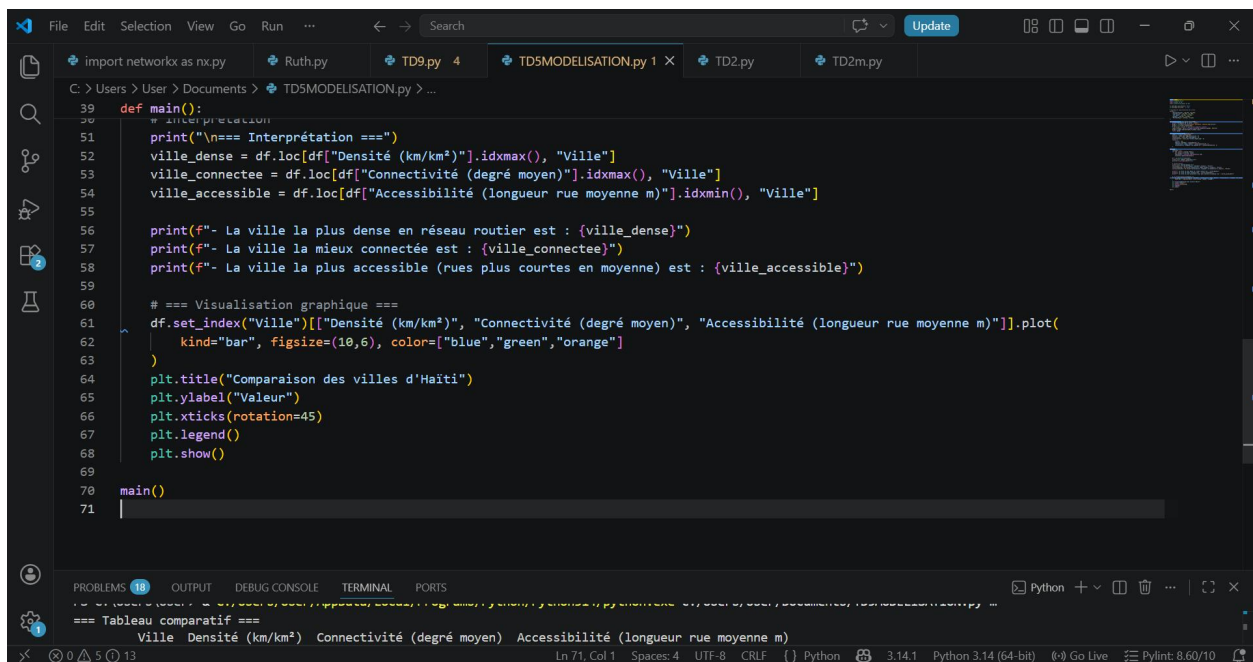
tableau comparatif

interpretation



```
1 import osmnx as ox
2 import networkx as nx
3 import pandas as pd
4 import matplotlib.pyplot as plt
5
6 ox.settings.log_console = False
7 ox.settings.use_cache = True
8
9 # Coordonnées approximatives des villes
10 villes = {
11     "Port-au-Prince": (18.544, -72.338),
12     "Cap-Haïtien": (19.759, -72.280),
13     "Les Cayes": (18.193, -73.746),
14     "Jacmel": (18.235, -72.535),
15     "Port-de-Paix": (19.933, -72.833)
16 }
17
18 def charger_ville(nom, point, dist=3000):
19     print(f"Chargement de la carte : {nom}")
20     graph = ox.graph_from_point(point, dist=dist, network_type='drive')
21     graph = ox.project_graph(graph)
22     # Nettoyage : garder le plus grand composant connecté
23     nodes_to_keep = max(nx.strongly_connected_components(graph), key=len)
24     graph = graph.subgraph(nodes_to_keep).copy()
25     return graph
```

Ville	Densité (km/km²)	Connectivité (degré moyen)	Accessibilité (longueur rue moyenne m)
Port-au-Prince	1.2	1.5	1.8
Cap-Haïtien	1.5	1.8	2.1
Les Cayes	1.8	2.1	2.4
Jacmel	2.1	2.4	2.7
Port-de-Paix	2.4	2.7	3.0



```
39 def main():
40     # Interpretation
41     print("\n=== Interprétation ===")
42     ville_dense = df.loc[df["Densité (km/km²)"].idxmax(), "Ville"]
43     ville_connectee = df.loc[df["Connectivité (degré moyen)"].idxmax(), "Ville"]
44     ville_accessible = df.loc[df["Accessibilité (longueur rue moyenne m)"].idxmin(), "Ville"]
45
46     print(f"- La ville la plus dense en réseau routier est : {ville_dense}")
47     print(f"- La ville la mieux connectée est : {ville_connectee}")
48     print(f"- La ville la plus accessible (rues plus courtes en moyenne) est : {ville_accessible}")
49
50     # Visualisation graphique
51     df.set_index("Ville")[["Densité (km/km²)", "Connectivité (degré moyen)", "Accessibilité (longueur rue moyenne m)"]].plot(
52         kind="bar", figsize=(10,6), color=["blue", "green", "orange"]
53     )
54     plt.title("Comparaison des villes d'Haïti")
55     plt.ylabel("Valeur")
56     plt.xticks(rotation=45)
57     plt.legend()
58     plt.show()
59
60 main()
61
```

Ville	Densité (km/km²)	Connectivité (degré moyen)	Accessibilité (longueur rue moyenne m)
Port-au-Prince	1.2	1.5	1.8
Cap-Haïtien	1.5	1.8	2.1
Les Cayes	1.8	2.1	2.4
Jacmel	2.1	2.4	2.7
Port-de-Paix	2.4	2.7	3.0

```
File Edit View Selection View Go Run ... Search Update
```

```
import networkx as nx.py Ruth.py TD9.py 4 TD5MODELISATION.py 1 X TD2.py TD2m.py
```

```
C:\> Users > User > Documents > TD5MODELISATION.py > ...
```

```
39 def main():  
40     ""  
41     # === Visualisation graphique ===  
42     df.set_index("Ville")[["Densité (km/km²)", "Connectivité (degré moyen)", "Accessibilité (longueur rue moyenne m)"].plot(  
43         kind="bar", figsize=(10,6), color=["blue","green","orange"]  
44     )  
45     plt.title("Comparaison des villes d'Haïti")
```

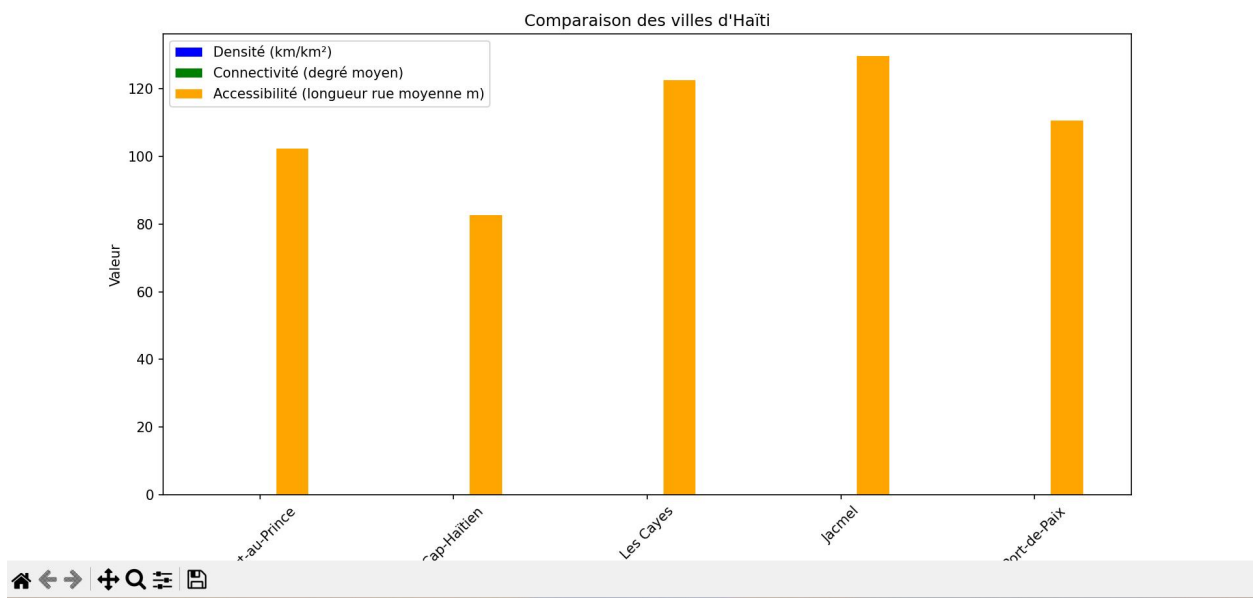
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [X] [Y] [Z]
```

```
PS C:\Users\User> & C:\Users\User\AppData\Local\Programs\Python\Python314\python.exe c:\Users\User\Documents\TD5MODELISATION.py  
Chargement de la carte : Port-au-Prince  
Chargement de la carte : Cap-Haïtien  
Chargement de la carte : Les Cayes  
Chargement de la carte : Jacmel  
Chargement de la carte : Port-de-Paix
```

```
=== Tableau comparatif ===  
Ville Densité (km/km²) Connectivité (degré moyen) Accessibilité (longueur rue moyenne m)  
Port-au-Prince 0 0 182.39  
Cap-Haïtien 0 0 82.64  
Les Cayes 0 0 122.60  
Jacmel 0 0 129.73  
Port-de-Paix 0 0 110.54
```

```
=== Interprétation ===  
- La ville la plus dense en réseau routier est : Port-au-Prince  
- La ville la mieux connectée est : Port-au-Prince  
- La ville la plus accessible (rues plus courtes en moyenne) est : Cap-Haïtien
```

Figure 1



TD5 -Optimisation de livraison

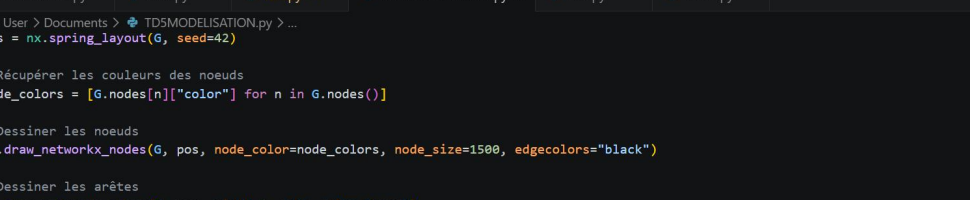
Objectif

Minimiser les trajets d'un livreur.

Travail demandé

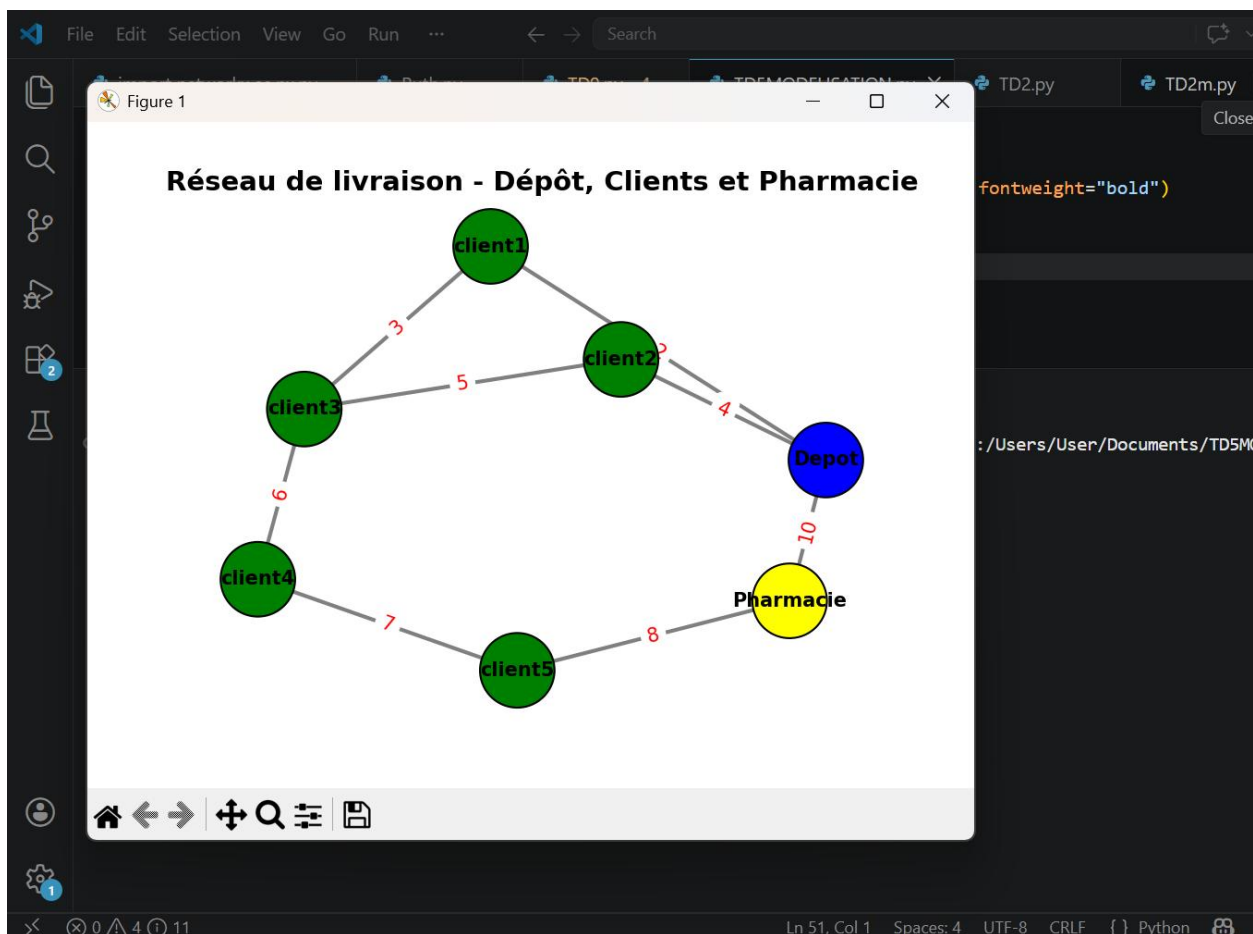
1. Définir plusieurs points de livraison
2. Trouver le meilleur parcours
3. Comparer methods

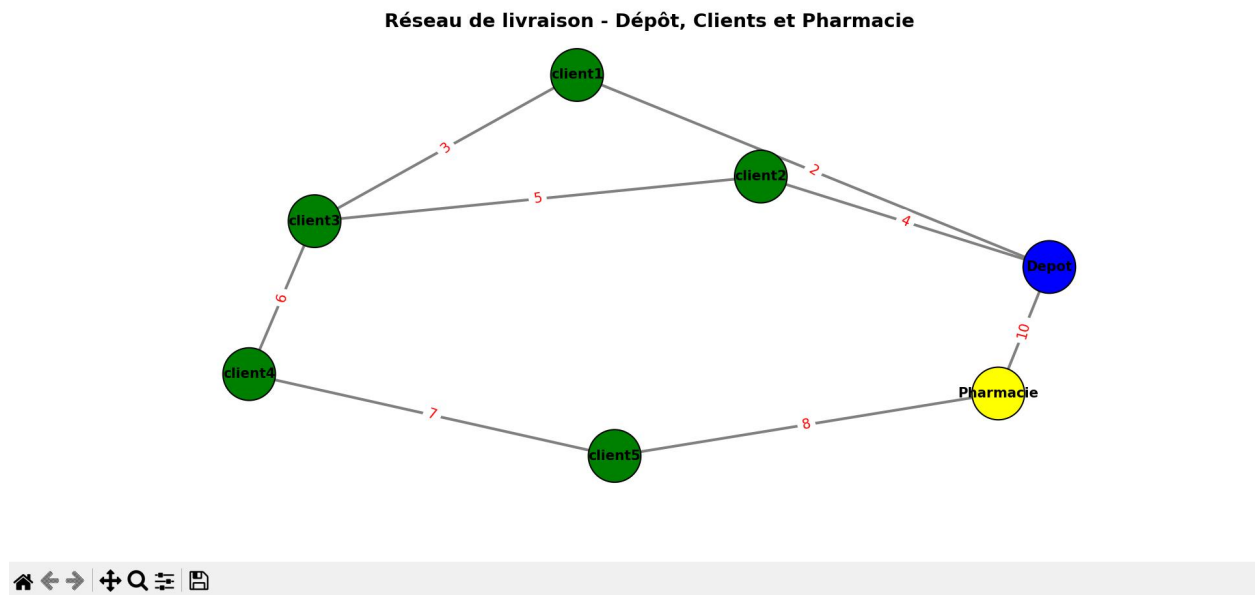
```
File Edit View View Tools Run ... Search
import networkx as nx
Ruth.py TD9.py 4 TD5MODELISATION.py X TD2.py TD2m.py
C:\Users\User> Documents > TD5MODELISATION.py > ...
1 import networkx as nx
2 import matplotlib.pyplot as plt
3
4 # Création du graphe
5 G = nx.Graph()
6
7 # Définir les noeuds
8 clients = ["client1", "client2", "client3", "client4", "client5"]
9 depot = "Depot"
10 pharmacie = "Pharmacie"
11
12 # Ajouter les noeuds avec attributs de couleur
13 for c in clients:
14     G.add_node(c, color="green")
15 G.add_node(depot, color="blue")
16 G.add_node(pharmacie, color="yellow")
17
18 # Ajouter des arêtes avec poids (distances fictives)
19 G.add_edge(depot, "client1", weight=2)
20 G.add_edge(depot, "client2", weight=4)
21 G.add_edge("client1", "client3", weight=3)
22 G.add_edge("client2", "client3", weight=5)
23 G.add_edge("client3", "client4", weight=6)
24 G.add_edge("client4", "client5", weight=7)
25 G.add_edge("client5", pharmacie, weight=8)
26
27 # Affichage du graphe
28 plt.figure(figsize=(10, 10))
29 nx.draw(G, with_labels=True, node_color=client_color,
30         edge_color="black", width=2, font_size=12)
31 plt.title("Graphique du réseau de livraison")
32 plt.show()
```



The screenshot shows the PyCharm IDE interface. The top toolbar includes icons for File, Edit, Selection, View, Go, Run, and a Search bar. Below the toolbar is a tab bar with several open files: 'import networkx as nx.py', 'Ruth.py', 'TD9.py' (selected), 'TD5MODELISATION.py', 'TD2.py', and 'TD2m.py'. The main editor window displays the code for 'TD5MODELISATION.py'. The code is a Python script that uses NetworkX and Matplotlib to create a network graph. It starts with importing 'nx' as 'nx'. Then, it creates a graph 'G' using 'nx.spring_layout' with a seed of 42. It iterates over the nodes of 'G' to assign colors to each node. Next, it draws the nodes and edges of the graph. The edges are drawn with a width of 2 and a gray color. Labels are added to the nodes and edges. The node labels are in bold font, and the edge labels are in red font. Finally, the graph is titled 'Réseau de livraison - Dépôt, Clients et Pharmacie' and is displayed using 'plt.show()'. The bottom status bar shows the current file path: 'PS C:\Users\User> & C:\Users\User\AppData\Local\Programs\Python\Python314\python.exe c:/Users/User/Documents/TD5MODELISATION.py'. The status bar also displays the current line and column (Ln 51, Col 1), the number of spaces (4), the encoding (UTF-8), the line endings (CRLF), the current file type (Python), and the version (3.14.1). The bottom right corner shows the Python interpreter (Python 3.14 (64-bit)) and the linter (Pylint: 8.57/10).

```
C: > Users > User > Documents > TD5MODELISATION.py > ...
29 pos = nx.spring_layout(G, seed=42)
30
31 # Récupérer les couleurs des noeuds
32 node_colors = [G.nodes[n]["color"] for n in G.nodes()]
33
34 # Dessiner les noeuds
35 nx.draw_networkx_nodes(G, pos, node_color=node_colors, node_size=1500, edgecolors="black")
36
37 # Dessiner les arêtes
38 nx.draw_networkx_edges(G, pos, width=2, edge_color="gray")
39
40 # Ajouter les labels des noeuds
41 nx.draw_networkx_labels(G, pos, font_size=10, font_weight="bold")
42
43 # Ajouter les labels des arêtes (distances)
44 labels = nx.get_edge_attributes(G, "weight")
45 nx.draw_networkx_edge_labels(G, pos, edge_labels=labels, font_color="red")
46
47 # Titre
48 plt.title("Réseau de livraison - Dépôt, Clients et Pharmacie", fontsize=14, fontweight="bold")
49 plt.axis("off")
50 plt.show()
51
```





Conclusion synthétique

Ces travaux montrent que l'analyse par graphes et la modélisation avec Python (NetworkX, OSMnx, Matplotlib) permettent :

- de **comprendre la structure des réseaux urbains**,
- de **comparer objectivement plusieurs villes**,
- et d'**optimiser des parcours logistiques** pour réduire les trajets.

En résumé, nous avons construit une approche complète allant de l'analyse théorique des réseaux routiers à leur application pratique dans la livraison et la mobilité urbaine.